MEGHANA RAJAKUMAR

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CREATIVE TECHNOLOGIST & ENGINEER, XR, APPLIED AI & REAL-TIME 3D

Extended Reality · Vision & Generative AI · Real-Time 3D & Technical Art · Full-Stack Prototyping · Cloud Automation

Creative technologist and engineer with 6+ years building XR, AI, and real-time 3D solutions at BMW Group and Epsilon (Publicis Groupe). I work across the full stack of making things real: modeling and animating 3D content, engineering interactions and tools in Unity and Unreal Engine, fine-tuning and deploying vision AI on edge hardware, building agentic and generative AI applications, and automating the cloud infrastructure underneath. My work includes a mixed reality digital twin featured on Meta's blog and covered by UploadVR, vision AI that enables 160+ real-time quality checkpoints on plant floors, and an agentic commerce app in ChatGPT now live in Europe. With a B.E. in Computers and Telecommunication Engineering and an MA in Animation and Interactive Technology from SCAD, I bring both engineering rigor and an artist's eye, and I enjoy leading ideas from first prototype to production alongside designers, engineers, and business teams.

CORE COMPETENCIES

XR & Real-Time 3D

- Enterprise XR & Mixed Reality
- Unity (C#) & Unreal Engine (5 yrs)
- Digital Twins & Simulation
- Modeling, Texturing, Rigging, Animation
- Procedural Tools (Houdini)
- XR Performance Optimization

AI & Computer Vision

- Vision AI & Video Action Recognition
- VLM & SLM Fine-Tuning
- Edge & On-Device Inference
- Agentic AI, RAG & Embeddings
- Synthetic Training Data
- Guardrails & OTel Observability

Engineering & Delivery

- C++, C, C#, Python, TypeScript
- React, Angular, FastAPI
- Tools, Plugins & Asset Pipelines
- Cloud (AWS, Azure) & CI/CD
- Agile Delivery & Architecture Reviews
- Partner & Stakeholder Leadership

Engines & Hardware: Unity, Unreal Engine, Meta Quest 3/Pro (Horizon OS), Meta Ray-Ban, NVIDIA Jetson
Creative Tools: Maya, ZBrush, Houdini, Blender, Adobe Substance 3D, Adobe apps, Pixyz, Figma
AI & Engineering: OpenCV, OpenGL, YOLO, NVIDIA NVILA, OpenTelemetry, Claude Code, Cursor, Codex, GitHub Copilot, Terraform, Ansible, gRPC

PROFESSIONAL EXPERIENCE

IT Technical Project Lead and Technical Artist

BMW Group, Greenville, South Carolina **Nov 2023 - Present**

- **Built 3D content, interaction systems, and the asset pipeline for Impossible Battery**, a mixed reality digital twin of BMW's fifth-generation high-voltage EV battery for Meta Quest 3/Pro, now used at BMW training centers in the US, Germany, South Africa, and China. Modeled, textured, rigged, and animated 70+ assets, built the hand-tracked tool interaction system in Unity (C#), and built a Pixyz-to-Unity CAD pipeline that automates ~80% of asset optimization. Demoed to partners including Meta; featured on Meta's blog and covered by UploadVR.
- **Owned Meta Quest 3/Pro testing and performance optimization**, profiling each build on both devices, reducing APK size while holding 90 FPS to prevent motion sickness, and turning client feedback into each next iteration.
- **Fine-tuned, evaluated, and deployed vision models for SmartEye Human Amplification**, BMW's Vision AI system that automates a 160+ step vehicle quality checklist previously done manually over days. Compared YOLO and vision-language models, generated synthetic training data in Unreal Engine, fine-tuned NVIDIA's NVILA, and deployed on-device inference on edge hardware, enabling hands-free QA while reducing latency and inference cost by 1.3x.
- **Led solution design and architecture reviews for XR and wearable AI programs**, guiding deployment decisions across Meta Quest, Meta Ray-Ban, and NVIDIA Jetson, on-device vs. cloud inference, and integration requirements for plant rollout.
- **Led the design and development of SmartEye Video Processor as primary contributor**, a self-service data preparation platform with a Python/FastAPI backend and Angular frontend that segments video into dataset groups, transcribes and labels content, and applies configurable transforms and anonymization to protect people on camera.
- **Contributed platform research, frontend optimization, and a capability framework to Customer Companion**, BMW's agentic commerce app in ChatGPT, now live in Europe, applying RAG, semantic search embeddings, and progressive profiling to personalize customer journeys.
- **Contributed to an AI-driven document intelligence platform now in pilot**, using AI agents to build knowledge graphs from interview video transcripts, building the React frontend, and working with the team on OpenTelemetry observability and strict grounding guardrails.

- **Built reusable developer tools**, including Unreal Engine plugins consumed by engineering teams across BMW's global development organization and Figma developer plugins for rapid prototyping of application frontends and workflows.
- **Served as primary technical point of contact for an external partnership with SCADpro**, guiding a partner team from concept and architecture review through production rollout with weekly demos and feedback cycles.

Software Engineer II and Cloud Automation Expert

Epsilon (Publicis Groupe), Bangalore, India **Jul 2019 - Sept 2022**

- Designed, implemented, and scaled infrastructure automation and deployment pipelines using Terraform, Ansible, and CI/CD frameworks, driving a 20% reduction in deployment times and a 32% improvement in deployment efficiency.
- Led complex cloud and production operations, including migrations across AWS and Azure, orchestration of critical workflows and data transitions, and major production release upgrades, minimizing downtime.
- Owned production reliability across tenants and supported hiring by interviewing entry-level technical candidates.

SELECTED PROJECTS

- **Impossible Battery (BMW):** mixed reality EV battery digital twin for technician training; see the project page on my portfolio for a detailed breakdown of my contributions.
- **Lego-fy (SCAD major project):** procedural Houdini tool that converts any 3D model into a LEGO-brick equivalent, built with Houdini, Unreal Engine, Maya, and Substance Painter.

EDUCATION

Master of Arts, Magna Cum Laude (3.7 GPA) **09/2022 - 06/2024**

Savannah College of Art & Design, Savannah, USA · Animation, Interactive Technology, Special Effects and Video Graphics

Bachelor of Engineering, Summa Cum Laude (4.0 GPA) **08/2015 - 07/2019**

RV College of Engineering, Bangalore, India · Computers and Telecommunication Engineering

ACHIEVEMENTS AND AWARDS

- **Gold Medalist in Engineering**, awarded for outstanding academic excellence and top performance in the program
- **Published researcher with 411+ IEEE citations**, bridging applied research and real-world technology commercialization
- **First place, Epsilon “Shark Tank” Gamechangers competition**, among 300 teams from 10+ nations
- **Interview Panelist** at the SCAD Faculty Conference, representing both BMW Group and SCAD; featured in the SCAD Events Blog